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T2D-MAS: From Software Ticket to Deployment

What the prototype does, how its agents work, and what the experiment shows.

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Agen Cerdas Enterprise · Project 1
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The project targets four delivery bottlenecks

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Original problem statement from the project report and idea document

1

Manual triage slows resolution

Severity, category and team routing depend on someone reading each ticket.

2

Duplicate reports create repeated work

The same incident is phrased differently, so keyword matching misses it.

3

Approvals become queues

Risky actions need a person, but reviewers rarely get the context they need.

4

Ticket quality varies

Reproduction steps, acceptance criteria and release notes are often incomplete.

Goal

Coordinate specialist software agents from user feedback to deployment, while people stay in control of high-risk decisions.

What is real and what is simulated

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The idea is a coordinated delivery team; the prototype runs all nine stages, some on simulators

1	2	3	4	5	6	7	8	9
Feedback capture real	**Triage** real	**Assignment** real	**Investigation** real trips, seeded logs	**Planning** simulated	**Implementation** simulated; secret scan real	**QA** simulated; NAPFD real	**Deployment** simulated; rules real	**Monitoring** simulated

Real: runs as built

Mixed: real trips, seeded data

Simulated: a seeded stand-in, labelled in the dashboard

Real

Feedback masking, duplicate search, triage and the autonomy gate, the Contract Net auction, signed trips to six data nodes, the six gates and the audit log.

Simulated with seeds

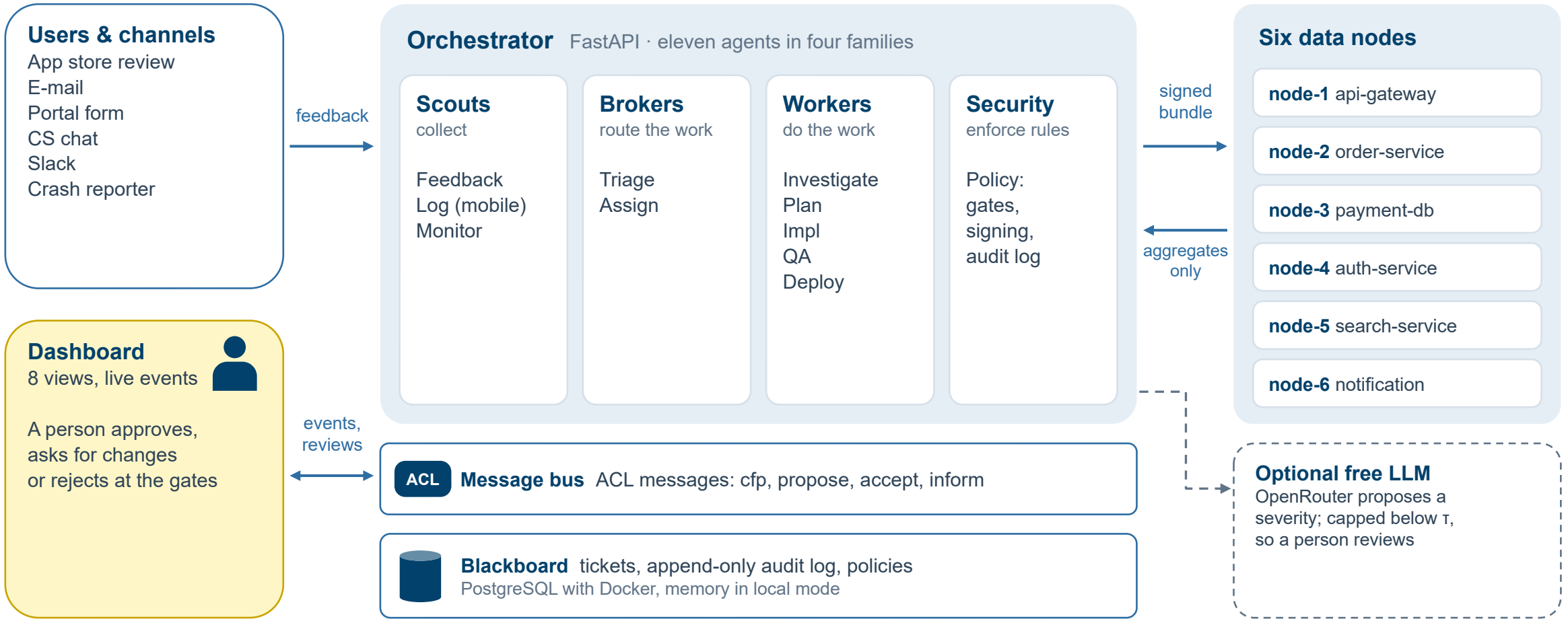
Node logs, project board, repository and draft change, CI, deployments and telemetry. Every panel in the dashboard says which it is.

OpenRouter (free models only) is optional for triage and reviewer notes. Its proposals can never approve a gated action, and the benchmark stays deterministic.

How the system fits together

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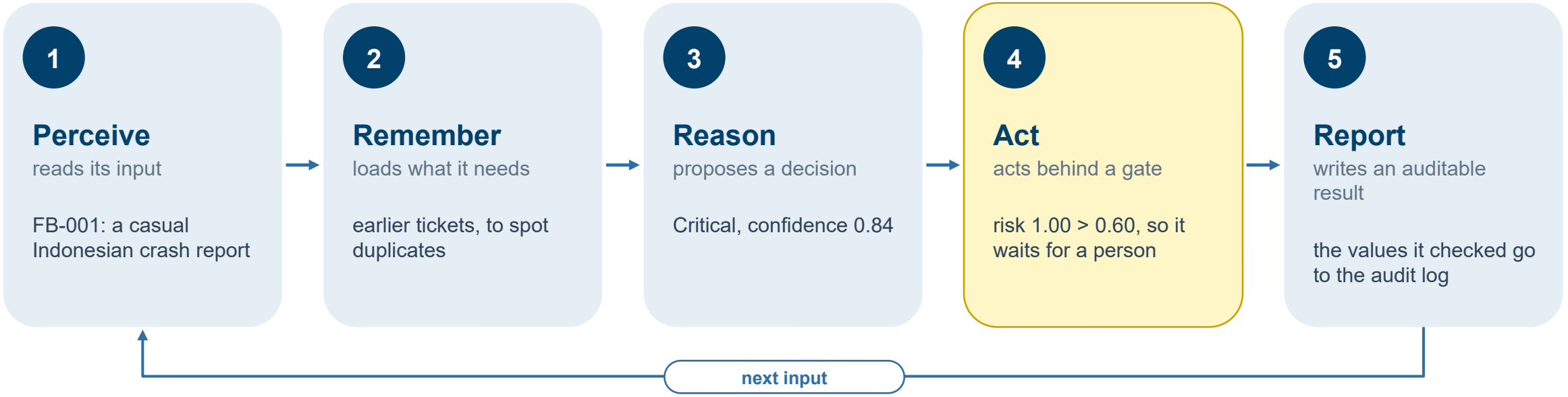
Feedback comes in, agents do the work, people decide at the gates, data stays on the nodes



Every agent runs the same five-step loop

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Shown for Broker-Triage on FB-001; the same contract applies to all eleven agents



Scouts
read feedback, visit the data nodes, watch releases

Brokers
check duplicates, judge impact, run the squad auction

Workers
investigate, plan, draft, test and release

Security
runs the six gates, where a person decides

A role says what an agent is responsible for; mobility says where it runs. Only Scout-Log travels.

A0–A3 change the coordination architecture

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All four arms use the same seeded synthetic tickets and five seeds

ARM	A0	A1	A2	A3
Architecture	Rules, one process	One agent	Multiple static agents	Multi-agent + mobile agent
Coordination	No agent messages	One agent loop	Agent messages between roles	Messages + signed bundle to node
Work location	Central process	Central process	Agents run centrally	Serialized mobile bundle at investigation
Measured mean bytes / ticket	0	1,778	3,556	4,161
Mean messages / ticket	0	6	12	13

The dashboard's Run a ticket path and its node trips are separate from the A0–A3 benchmark.

Run a ticket: nine agents from feedback to release

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Real decisions and gates; the repository, CI, deploys and telemetry are seeded simulators

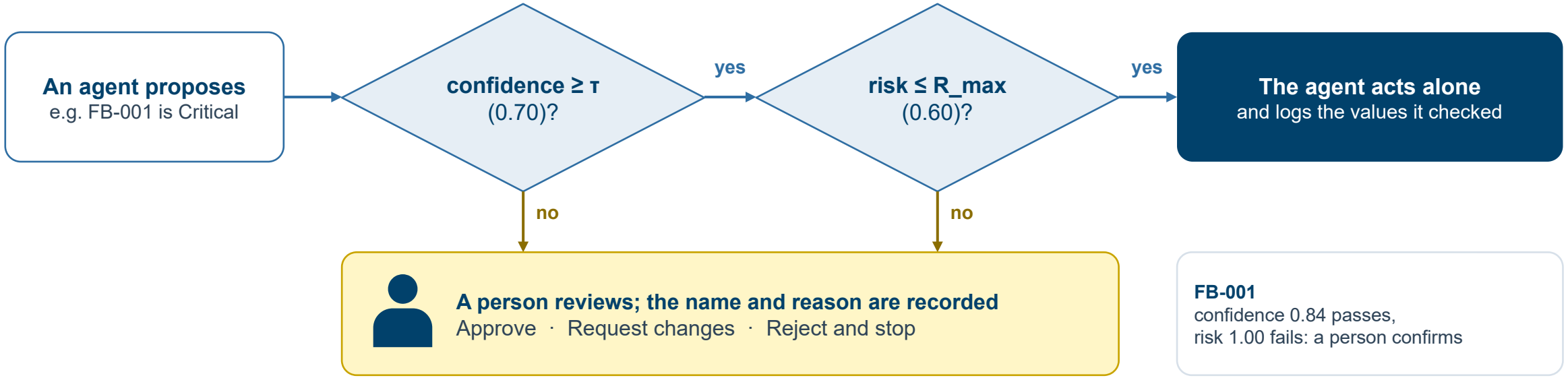


Merge and release always wait for a person; four more gates stop only when their rule fires.

When an agent may act alone

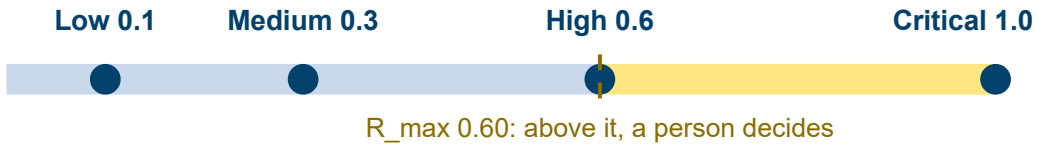
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Two checks at every decision; if either fails, a person decides



FB-001
confidence 0.84 passes,
risk 1.00 fails: a person confirms

Risk comes from the severity



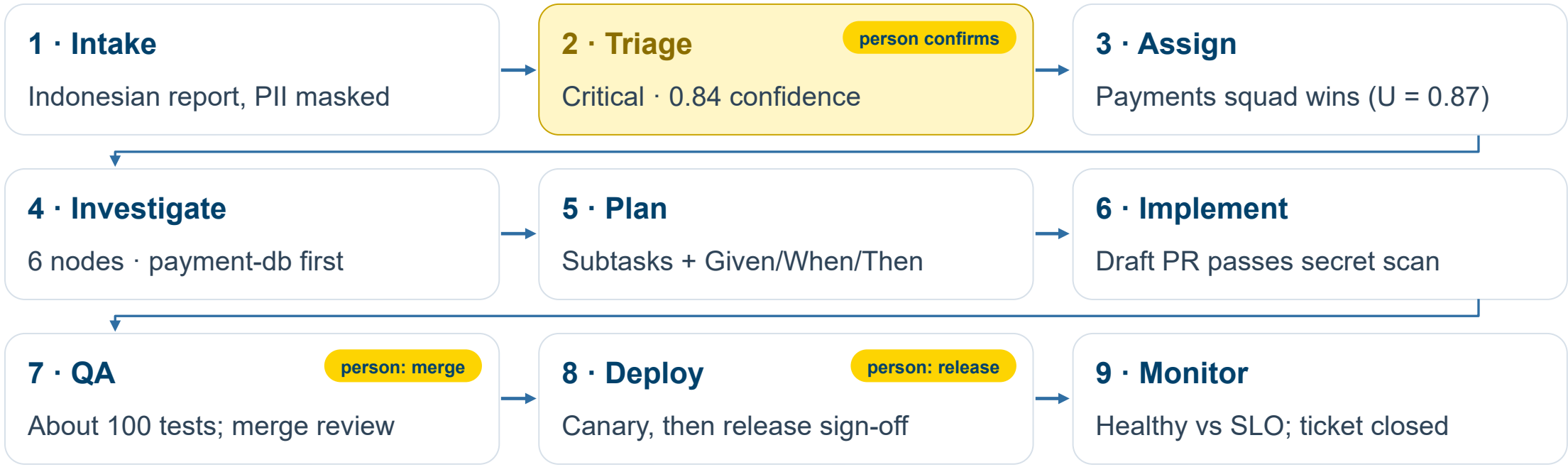
Six gates

Always a person: merge review, release sign-off
Only when a rule fires: duplicate 0.70–0.85, autonomy gate, secret found, canary risk > 0.30

Example: one flash-sale complaint, end to end

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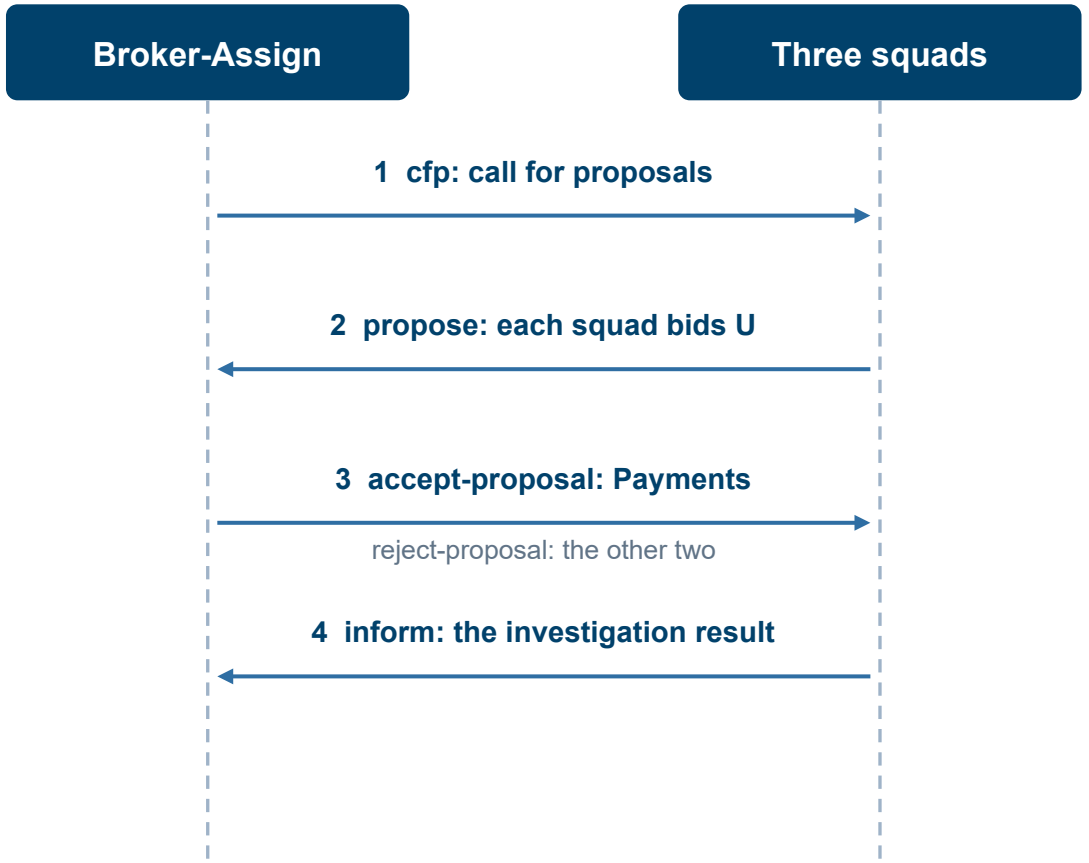
FB-001, a casual Indonesian crash report, waits for a person at triage, merge and release



Critical risk (1.0) is above R_max = 0.60, so triage waits for a person; merge and release always do. A repost of the complaint (FB-002) scores 0.956 and is auto-linked.

Contract Net: the squads bid for the ticket

Broker-Assign announces the task, and the best bid wins the investigation



FB-001: the bids



U = 0.5·skill + 0.3·(1 − load) + 0.2·(1 − cost)

Payments: 0.5·0.92 + 0.3·0.90 + 0.2·0.70 = 0.87

The winner investigates. The SLA follows the severity: Critical means 2 hours.

AI integration is hybrid, with a bounded optional LLM

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The interactive path, benchmark and separate ML modules use different methods

Live in the dashboard

Severity: English and Indonesian rules; an optional free OpenRouter model may propose, with confidence capped at 0.69

Duplicates: TF-IDF cosine over character n-grams

Assignment: the Contract Net auction

Root cause: GraphSAGE-mean ranking over the call chain

Reviewer notes: the optional model can structure a note; a person decides

In the benchmark (A0–A3)

LLM: none; every run is seeded and deterministic

Severity ROC: TF-IDF with a calibrated LinearSVC

Duplicates: a word-overlap stand-in for Recall@5 and MAP

Test order: scored with NAPFD

Offline modules (trained, not live)

Duplicates: MiniLM embeddings with FAISS

Severity: DistilBERT

Root cause: trained GraphSAGE, GNNExplainer, PageRank

CI and canary: PPO and DQN policies

Models propose; people decide. No model writes code, and CI and deploys are seeded simulators.

A2 vs A3: static and mobile execution

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Mobility is about moving a signed agent to the data, not adding another role

A2 · static agents

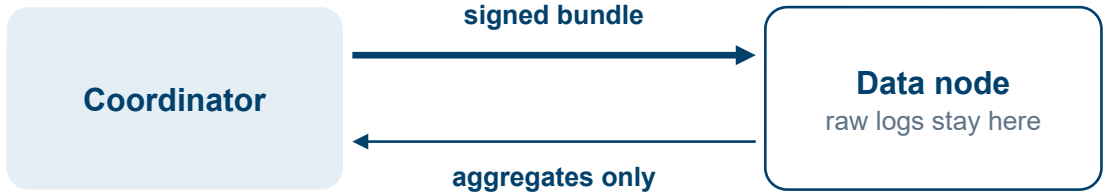
Agents run centrally. To investigate, the coordinator pulls data or evidence from the node.



Moves the data: heavy when the logs are large.

A3 · with a mobile agent

A signed bundle runs at the node. The node keeps its raw data and returns allowlisted aggregates.



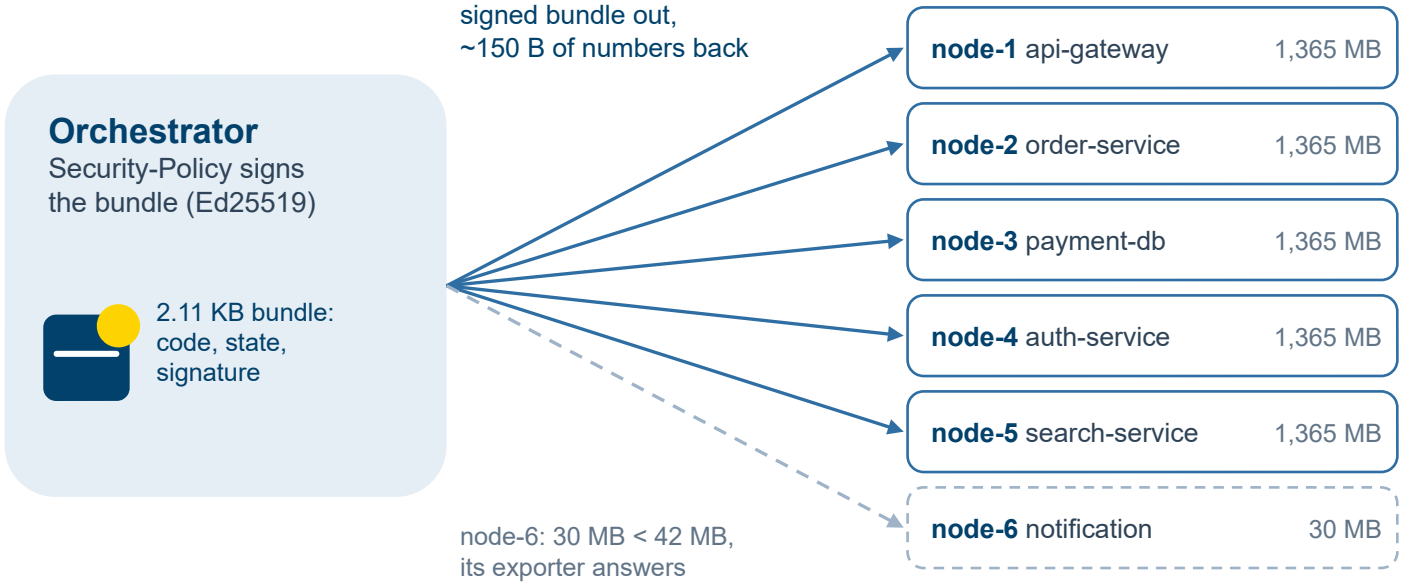
Moves the code: pays off above about 14 MB per node.

In the benchmark, A3 sends 4,161 bytes per ticket against 3,556 for A2: the generated logs are tiny, so the bundle is overhead. In the dashboard, Scout-Log makes these trips during investigation (next slide).

Scout-Log: moving the code to the data

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The mobile agent visits six data nodes; the raw logs never leave them



1 Sign
the key never leaves the orchestrator

→

2 Travel
one hop to each node

→

3 Verify, run
isolated process, 20-second limit

→

4 Return
allowlisted numbers only

Static pull vs mobile agent

Report cost model, TCK-1042: 6 nodes × 1,365 MB

Static pull: every raw log to one server

8,190 MB · ≈ 10 min 55 s

Mobile agent: the code goes to the data

84 MB · ≈ 27 s

98.97% fewer bytes, about 24× faster

Move only if the node holds more than
(12 + 2) × 3 = 42 MB of logs

Live trip in the demo
11.5 KB moved; 6,855 MB of raw logs stayed on the nodes

The benchmark result set is synthetic

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The repository can load public bug data, but the reported A0–A3 experiment used generated tickets

A0–A3 benchmark

500 tickets per arm: 100 generated tickets × 5 seeds (42–46). Version: synthetic-eval-v1. The same generated cases feed all four arms.

Ticket fields

ID · title · body · component · severity · creation time · duplicate link · fault label

Optional source datasets

Loaders support GitBugs and Eclipse/Mozilla issue exports. Preprocessing masks common PII patterns, collapses repeated stack-trace lines and creates chronological 70/10/20 train, validation and test splits.

Interpretation

Benchmark split: first 70% train, final 30% evaluate. General data loader split is 70/10/20. DORA-style outcomes are simulated; token use is estimated.

A1–A3 improve synthetic retrieval and fault ordering

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Run cited in the report · 500 generated tickets per arm · host timings are only workload proxies

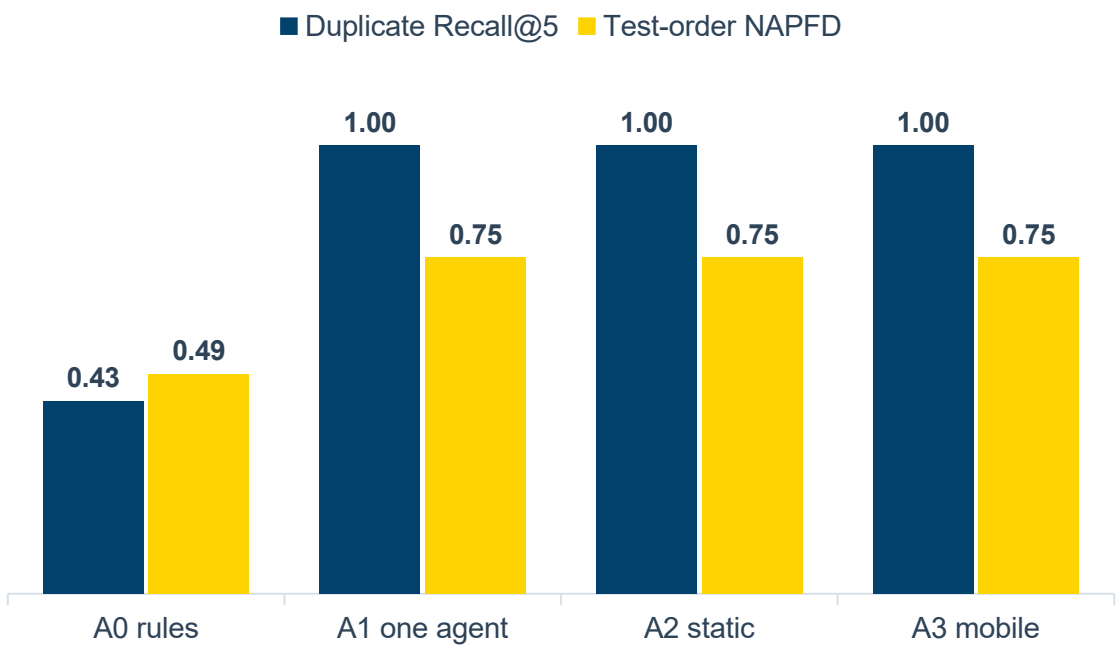
Metric	A0	A1	A2	A3
Mean wall time / ticket	0.102 ms	0.155 ms	0.184 ms	0.198 ms
Network bytes / ticket	0	1,778	3,556	4,161
Duplicate Recall@5	0.429	1.000	1.000	1.000
Duplicate MAP	0.242	0.680	0.687	0.685
Severity macro-F1	1.000	1.000	1.000	1.000
Fault ordering NAPFD	0.490	0.750	0.750	0.750
Simulated rollback rate	5.6%	5.4%	5.8%	7.4%

A3 adds the most bytes and has the highest rollback rate. All arms reach F1 = 1.0 on distinctive synthetic labels.

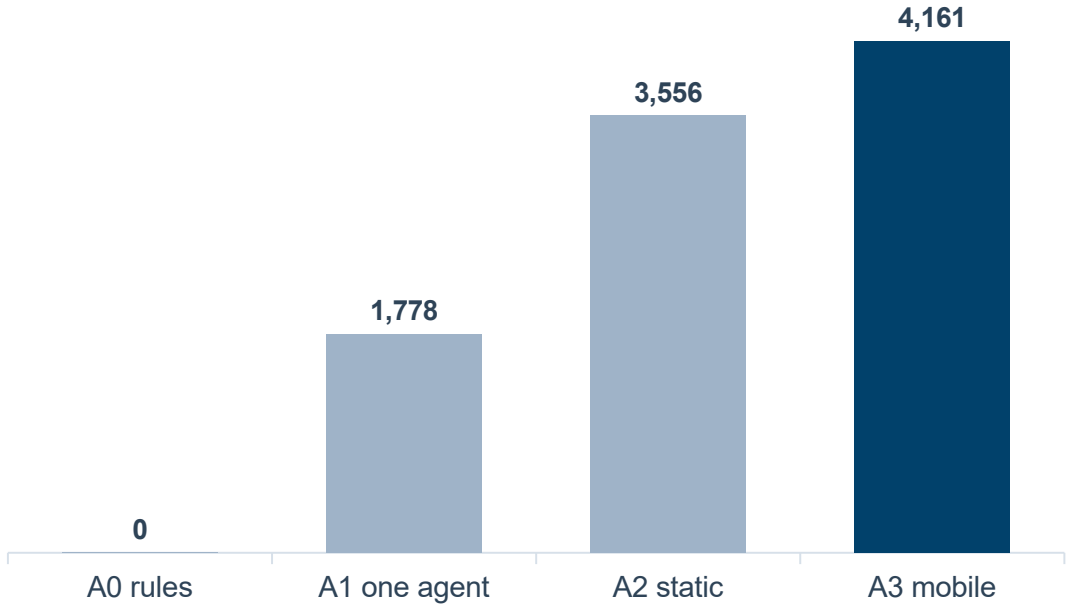
Results at a glance

The run the report cites: 500 generated tickets per set-up, five seeds

Quality (higher is better)



Cost: data sent per ticket (bytes)



Agents beat the rules at finding duplicates and ordering tests. Several agents do not beat one on accuracy. The mobile bundle adds bytes on tiny logs.

Treat these as classroom results, not field evidence

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Seeded results support discussion of trade-offs; they do not validate business impact

0.0018

Friedman p, data sent

The set-ups differ; A3 sends the most bytes.

0.0136

Friedman p, wall time

Tiny CPU-proxy timings, not service latency.

0.604

Friedman p, rollback

Not significant; neither is CPU ($p = 0.926$).

20 / 20

violations blocked

No full release without approval; A3 had 11 approved simulated releases.

What the result says

Coordination and mobility add serialized work in this set-up. Whether running the analysis next to the data is worth that cost depends on data locality and privacy, which this benchmark does not measure.

No field productivity claim

The harness runs no hosted LLM, Redis, real CI or production deployment, and its DORA-style outcomes are simulated.

Conclusion: a useful prototype, not autonomous delivery

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The implemented boundary is clear, and it gives the next evaluation a practical target

What works today

Nine agents take natural-language feedback to a monitored release: live duplicate search, a squad auction whose winner investigates, signed node trips, six gates and a monitoring loop.

What the experiment shows

Agent arms improve duplicate retrieval and fault ordering in generated data. A3 adds transfer cost; the interactive dashboard is a different path, and real data locality is not measured.

Conclusion

T2D-MAS shows inspectable agent collaboration with people in control. Next, replace the simulators with a real repository, CI, release pipeline and telemetry, move the offline models into the live path, and evaluate on public data.

Keep the boundary clear: models propose, humans decide, irreversible actions remain gated.

Sources and scope notes

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Repository and supplied project documents checked for this deck

- Project 1 - Idea.pdf — original problem, illustrative cosine example, agent design and AI capability roadmap.
- Laporan_Proyek1_T2D-MAS_Final.pdf — expanded design, agent taxonomy, methods, intended workflow and the cited results (§9.4–9.5).
- README.md; docs/AUDIT.md; docs/process_and_experiments.md; agents/AGENT_CATALOG.md — architecture, pipeline and what is real, simulated or not built.
- agents/pipeline.py; scouts.py, brokers.py, workers.py; policy.py; severity_triage.py; deliverables/T2D-MAS-Live-Demo-Guide.md — the nine agents, their gates, optional OpenRouter and the demo walkthrough.
- eval/runner.py; eval/config.json; results/20260930T020846305626/summary.csv, statistical_tests.csv and security_injection_summary.csv — the benchmark outputs the report cites.
- data/README.md; triage/README.md; rca/README.md; envs/README.md — dataset loaders and separate model/simulation modules.

A fresh run matches the cited CSVs on every non-timing value (eval.compare_runs). Timing, simulation and estimate labels remain explicit.

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Thank you.

Code, data and results:

github.com/zidanarsyad/t2d-mas

